BNPDFormer Best Checkpoint Metrics

`dense_i1_a1_prefix8` (unsupervised, epoch 8)

Official retained recipe: `FactoryBN_dense_f1_p80.json`

Protocol. Dataset `dense_i1` (N=38, F=27); episode split 140/28/36 (seed 42); windows 20826/4331/5354. Input T_{in}=30 min, horizon H=15 min. A report is correct iff the station matches *and* $|\hat{s} - s| \le 3$ min. Thresholds are selected on validation with report precision ≥ 0.80, then frozen on test. Occupancy-grid F1 and IoU-event F1 are not checkpoint metrics and are omitted (not recorded in the retained notes).

Table 1: Station-level event reports (primary task A.1). Checkpoint metric is report F1.

Group	Metric	Validation	Test
Report (primary)	Precision P	0.883	0.852
	Recall R	0.881	0.856
	F1	0.882	**0.854**
Stratified recall	Upcoming R_{up} ($s > 0$)	0.633	0.455
	Ongoing R_{on} (s=0)	0.920	0.911
Support	n_{true} upcoming	98	112
	n_{true} ongoing	624	828
	n_{true} total	722	940

Table 2: Order remaining time and process-cause heads (do not override report F1).

Layer	Metric	Validation	Test
Order remaining time	MAE(*remain_len*) (min)	≈ 12.0	**10.5**
Process cause (6 classes)	Overall accuracy	—	0.965
	Macro recall (support>0)	0.774	**0.971**
	Majority-class baseline (train)	0.41	
	Valid samples n_{cause}	—	579
Per-class recall	*transport_delay*	—	0.956
	material_shortage	—	0.988
	starved_upstream	—	0.949
	queue_buildup	—	0.992
	blocked_downstream	no support on test	
	high_utilization	no support on test	

Notes. (1) Taken from retained notes for `prefix8`; `last_metrics.json` is not in this workspace, so `who_*`, `start_mae`, `dur_mae`, occupancy-grid F1 and IoU-event F1 are not invented. (2) Cause labels are process-heuristic: report test macro 0.971 together with val macro 0.774 and the train majority baseline 0.41; this is not a causal claim. (3) Upcoming recall 0.455 (112 test events) remains the open early-warning failure mode. (4) Rejected on the same protocol: precursor head (F1 0.827), score regression (0.850), STGNPP auxiliary (0.849).